6.1 – Heat/Temperature/Energy

1. Define the following terms
   Energy:
   Temperature:
   Heat:

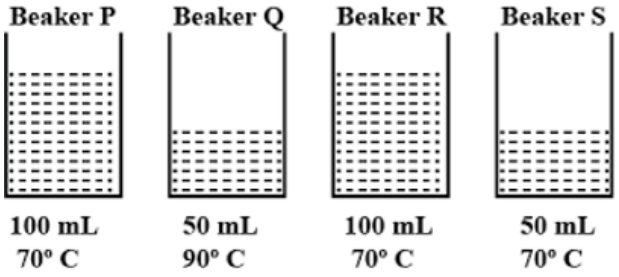
2. Discuss how a substance transfers energy. Think about what occurs at the molecular level.

3. Describe the difference between an endothermic and exothermic process.

4. Write out 2 scenarios where one is an endothermic process, and one is an exothermic process.

5. Use the diagram to the right to answer parts a through …
   a. Which beaker contains particles with the highest average kinetic energy. _____

   b. Describe the average kinetic energy of beakers P, R, and S.



6. A student places an ice cube on their hand. Describe the scenario and use the following terms in your description: energy, temperature, system, surroundings, endothermic OR exothermic. Draw a diagram to support your explanation.

7. 0 Kelvin is the temperature at which "absolute zero" is achieved. Describe the behavior of the molecules at this temperature.

Topic 6.2 - Specific Heat Calculations

1. Define specific heat. _____

2. Specific heat equation is _____.

3. What does each variable in the specific heat equation represent?

Use the table on the right to answer the following questions.

4. Which metal requires the least energy to raise its temperature? The most energy?

5. If you heated 2 grams of each material at the same temperature, which metal would be the hottest after 10 minutes?

Element	J/g℃
Aluminum	0.9012
Tin	0.2167
Magnesium	0.4686
Lead	0.1289
Zinc	0.3883

6. If 200 grams of water is to be heated from 24.0°C to 100.0°C to make a cup of tea, how much heat must be added? The specific heat of water is 4.18 J/g·C

7. The specific heat of wood is 2.03 J/g·°C. How much heat is needed to convert 550.0 g of wood at -15.0°C to 10.0°C?

8. What is the specific heat of a substance that absorbs 2505 joules of heat when a sample of 100 g of the substance increases in temperature from 10 °C to 70°C?

9. How many grams of water would require 2230 joules of heat to raise its temperature from 34.0°C to 100.0b°C? The specific heat of water is 4.18 J/g·C

10. A block of aluminum weighing 140.0 g is cooled from 98.4°C to 62.2°C with the release of 1080 joules of heat. From this data, calculate the specific heat of aluminum.

11. A 55 kg block of metal has an original temperature of 15.0°C and 0.45 J/g·°C. What will be the final temperature of this metal if 450 J of heat energy are added?

12. Granite has a specific heat of 800 J/g·°C. What mass of granite is needed to store 150,000 J of heat if the temperature of the granite is to be increased by 15.5°C?

Specific Heat of Ice	2.11 J/g°C
Specific heat of Liquid Water	4.184 J/g°C
Specific Heat of Water Vapor	2.00 J/g°C
Enthalpy of Fusion for Water	6.01 kJ/mol
Enthalpy of Vaporization for Water	43.9 kJ/mol

6.3 – Heating and Cooling Curves

1. Indicate the states of matter that exist at points A, C, and E

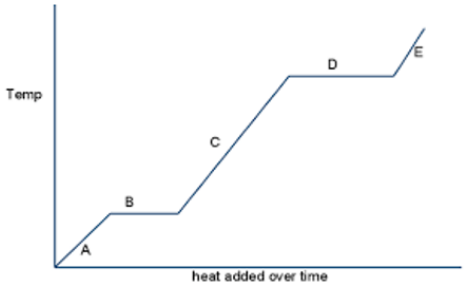
A: _____

C: _____

E: _____

2. A student heats 2.5 grams of ice from -15°C to -5°C. The student sets up the following equation to solve for q. Is the student correct? If not, describe the change the student needs to make.

$$q = (2.5\ g) * (4.184\ J/g^oC)*(-5^oC - (-15^oC)$$



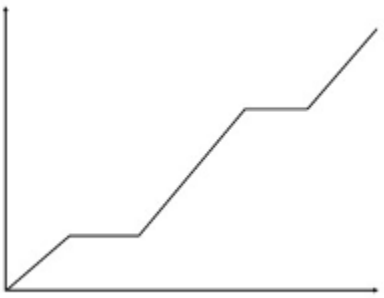
3. What processes occur at position B on the graph? Why does the temperature not change during this process?

4. What processes occur at position D on the graph?

5. Describe enthalpy of fusion and what the value represents. What position on the graph indicates the point when the enthalpy of fusion value is needed in your calculations?

6. Describe enthalpy of vaporization and what the value represents. What position on the graph indicates the point when the enthalpy of vaporization is needed in your calculations?

7. A 25.0 gram sample of water is heated from 75.0°C to 115.0°C. Calculate the amount of energy needed to perform this process.



a) How many q values do you need to calculate? Indicate each q value on the blank graph provided. Label as q_1, q_2, q_3, etc…

b) Write out each equation below that is needed to solve the problem. Calculate each q value.

c) Calculate the total amount of energy, in kJ, needed to complete this process.

8. 15.0 grams of water is cooled from 105.0°C to 85°C. Calculate the total change in energy, in kJ, for the process occurring. Similar math to what was done in question 7. One note though: you are cooling the water down. Consider how it affects your equations!

Topic 6.4 - Thermochemical Equations and Calculations

1) What is the enthalpy of a reaction?

2) Use the equation below to answer parts a through d.

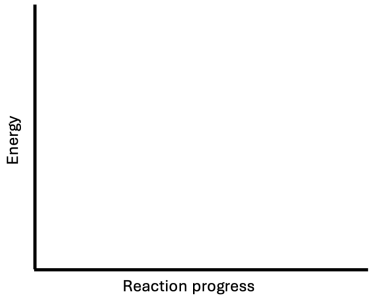
___C_3H_8 + ___O_2 → ___CO_2 + ___H_2O ΔH = -2043 kJ/mol rxn

a) Is the reaction endothermic or exothermic? Justify your answer.

b) Calculate the energy released if 0.150 moles of oxygen gas is combusted in the presence of excess propane

c) How many grams of propane were combusted if 3525 kJ of energy were released in a reaction.

d) On the axis provided, draw a reaction profile diagram for the combustion of propane. Label the parts of the diagram.



3) Use the equation below to answer parts a through e.

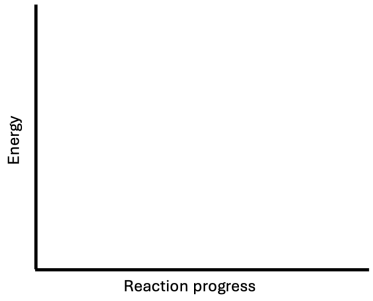
___CO_2 + ___H_2O —> ___$C_6H_{12}O_6$ + ___O_2 ΔH = 2814 kJ/mol rxn

a) Is the reaction endothermic or exothermic? Justify your answer.

b) How many grams of CO_2 are reacted if the reaction absorbs 755 kJ of energy?

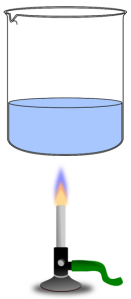
c) The above reaction represents photosynthesis. A mature oak tree can produce 274 liters of oxygen gas a day. How much energy, in kJ, is absorbed by the tree to produce that much oxygen per day?

d) On the axis provided, draw a reaction profile diagram for the photosynthesis reaction. Label the parts of the diagram.



Topic 6.5 - Calculating Enthalpy of Reaction Through Calorimetry

1) A student performs an experiment where the combustion of methane is used to heat a beaker full of water. A diagram is provided below that illustrates the setup of the equipment. In the experiment, we are assuming that all energy is transferred between the reaction and the beaker of water. No heat is lost to the air. Use this experiment to answer parts a through …



___CH_4 + ___O_2 —> ___CO_2 + ___H_2O ΔH=___ kJ/mol rxn

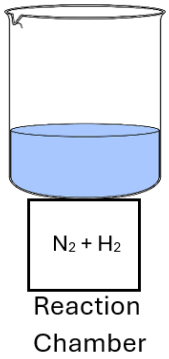
a) The student burns 3.75 grams of methane (CH_4) with excess oxygen gas. Calculate the mol rxn.

b) Calculate the energy change of the water. ($q_{surroundings}$)

c) Find the enthalpy of the reaction in kJ/mol rxn

Mass of water in beaker	450.0 grams
Initial Temperature of Water	20.0°C
Final Temperature of Water	89.0°C

2) A student performs an experiment according to the diagram to the right. Nitrogen and hydrogen gas are combined in a chamber and reacted below a beaker of water. Assume that energy is being transferred between the reaction chamber and the beaker of water. Assume that no energy is lost to the air.



Reaction: ___N_2 + ___H_2 → ___NH_3 ΔH = -46.0 kJ/mol rxn

a) 5.0 grams of hydrogen gas are reacted with XS nitrogen gas. Calculate the energy of the reaction and label it as endothermic or exothermic.

b) Assuming that all the energy is transferred to the water. Calculate the final temperature of the water if there are 250.0 grams of water at an initial temperature of 25.0°C.

c) How many grams of hydrogen would need to be reacted to raise the temperature of the water to its boiling point starting at an initial temperature of 25.0°C?

Topic 6.6 - Extra Practice (Putting it all together)

1. Knowing that the water melts at 0°C and boils at 100°C, how much energy must be added to 12.0 g of ice at -15°C to form steam at 115°C? The specific heat of liquid water is 4.18 J/g^{0}C. The specific heat of gaseous water is 2.03 J/g^{0}C. The specific heat of solid water is 2.09 J/g^{0}C.The heat of fusion is 2260 J/g. The heat of vaporization 2,260 J/g°C.

2. Explain why an increase in pressure will affect the boiling point of a liquid.